Case Study: Nxt Day Business Mail

Summary

Nxt is an independent provider of early morning, next day business mail (Business to Business). It consists of two companies, one specialising in non-secure items (Nxt Service) and one specialising in secure items (Nxt Secure) under a single Group Chief Executive (The Nxt Day Group). The Group handles more than 27m items per year. The two companies operate independently but share a network infrastructure comprising of a joint network with 2,500 employees including delivery staff. They have 1,200 delivery vehicles in their fleet. Where possible both companies' sales teams refer business to each other. It has not been possible to fully merge the two businesses because not all Nxt Service sites are sufficiently secure for the Nxt Secure operation.

The company delivers 75% of all items through sub-contract courier companies. A contract courier company sometimes collects, but usually delivers items shipped through the Nxt system under Nxt's terms and conditions at a fixed price. The Couriers feed delivery confirmation data for Nxt back via collection signature cards that are returned to Nxt and scanned for customers to query on the next working day. Records are kept by Nxt using bar-coded labels applied to packages when items arrive at Nxt's outward distribution depot.

Currently there are no formal performance measures for subcontractors, but any complaints or anomalies are recorded by Nxt's call centre, and fed back to the sub-contractors. The sub-contractors may keep internal digital records, but they do not co-ordinate these with Nxt.

(See attached diagram for the sequence of delivery and reporting events)

Financial Summary

The Group has combined annual revenues of £100 m (approx. 50/50 split between Nxt Service and Nxt Secure). Both companies are marginally profitable and have limited internal finances to invest.

Strategy

Nxt Service covers 204k business postcodes (33% of all businesses in the UK). Data on delivery points is maintained using an externally sourced Postcode Address File (PAF). This database is used throughout the UK. Nxt target small professional service companies (such as legal, property, travel and recruitment) and target additional businesses in close proximity to existing postcodes that they serve.

Nxt would like to integrate their systems with data from PAF to use in a marketing information system, which may require a new system that can interface with PAF to record sales leads, and track sales volumes, customer by customer, postcode by postcode, and region by region. Nxt needs to be able to track their own performance, as well as the quality and delivery performance of their sub-contractors.

Quality is defined by the speed of delivery, the courtesy of the couriers, and the condition of the packages that arrive at a customer's premises. Performance is measured by the sub-contractors ability to liaise with Nxt, the accuracy and reliability of delivery times and that they meet the product specification (i.e. Delivered by 9:00 am or 12:00pm) and also the collection of a delivery signature. This includes the return of that signature to Nxt. This relies heavily on the information provided by the subcontractor, unless a customer makes Nxt aware of a problem not flagged up by the subcontractor.

Nxt Day Group has been working with an external IT company to look at ways to use software to analyse traffic and maximise use of its own delivery network (currently delivering 25% of all items) as well as to improve the quality of information provided to customers. Currently there is an asset database which records all plant, fleet and IT assets, but it does not track how these assets are used. It is keen to be able to track these resources by location, use, and cost. Improving this system will give many benefits to Nxt, and tangible benefits to customers by making deliveries more efficient, and making it easier for Nxt to use these resources to best advantage. This will also allow Nxt to define and formalise processes so as to be able to more accurately predict performance, and take corrective measures.

Whilst Nxt does have a system that schedules, manages, and monitors delivery, it may not work with the proposed changes. The new system will need to interface with the customer and asset databases, which will likely require the acquisition of new IT assets to facilitate the integration of new technology. It will also require additional functionality for the software, which needs to be taken into consideration.

Nxt Service operates a limited Business to Consumer service for key customers using Nxt Secure for final delivery. These services require a secure facility, and more robust security while handling customer packages. Signatures of named people are always required, so the database must be able to store named, authorised signatures for the release of secure packages.

Customers are frequently unable to confirm the delivery of a package until the next business working day as the system relies on the return and management of signature cards from the Nxt fleet and subcontractors. This is a particular problem in a case where a delivery hasn't been made. It makes resolving the issue more costly and time consuming for everyone. In the very fluid courier marketplace, this can lose Nxt customers. This is amplified if an error is made by a sub-contractor as they sometimes do not hand in cards until they come to the depot on the next day to pick up their next batch of delivery items.

In all cases, Nxt needs to be able to maintain a billing and customer receipt system, ideally one that can be accessed online. The existing system is dependent on the receipt of signatures, and other manual authorisation processes. Such delays to billing can be difficult for customers and Nxt- if customers are not billed in a timely manner they may not pay in a timely manner. As a result, cash-flow may be affected, especially in a business where so many deliveries are made by contractors who also rely on cash-flow to keep operating.

Billing needs to be fully automated, and able to interact via sub-contractor reporting systems (where they exist). This may mean an online solution, in which sub-contractors can enter data directly into the Nxt database. Customers would ideally be able to tell right away when a delivery is made on-line or perhaps by e-mail for customers using premium services. Such a system will enable Nxt to negotiated better prices with customers and improve cash-flow.

Leadership and Staffing

The business has had a relatively traditional management structure for many years, but the management team was recently restructured to reflect the needs of a changing business climate. The management structure now consists of 5 managing directors reporting to the Group Chief Executive.

Management Structure

The senior management team is seen as highly capable and the staff structure is stable. The Group CEO is seen as charismatic and an effective leader having previously been MD of Nxt Secure whom he helped found in 1990.

Nxt Service and Nxt Secure aim to continue to focus on their niches for the next 12 months but will also identify potential opportunities to develop new shared services.

Nxt Day Group, Nxt Service and Nxt Secure are all headquartered in Wolverhampton.

Their biggest market is in Central London where they have their own delivery staff who are non-uniformed and non-unionised. However, nationally 75% of all deliveries are undertaken by local / regional courier companies on a flat rate payment per drop basis.

Products and Services

Nxt Group delivers approximately 1/2 million items per working week and customers can 'post' by taking their mail to any one of 4500 'mail exchange' sites (typically in the foyers of large shared occupancy offices) or arrange for a Nxt vehicle to call to collect direct.

Sales

Nxt Group are competitive with other mail service providers and offer discounts on annual spend above £25k. They negotiate price changes annually with customers. This process is currently manual, with each transaction agreed individually by the customer and sales force, but for consistency and control purposes, Nxt would like to automate the discount process. This would mean having a customer database, and it must be able to identify customers by groups – based on profitability, trust (do they pay their invoices in a timely manner?) and volume.

Network Capability

The Group operates a hub and spoke distribution network with a national hub near Coventry and additional regional hubs serving 25 sorting depots that link to the local 'mail exchange' sites or serve customers direct. The regional hubs are in Cardiff, Edinburgh, Belfast, Manchester and Watford.

The 25 sorting depots are equipped with a limited automated sorting capacity that 'reads' addresses and sorts items using PAF to the nearest depot. Sorting information is not currently stored for practical reasons.

Target Sectors

Having initially focused on the legal, professional, health, property and travel sectors, Nxt are now targeting the public sector, local authorities and new business parks. The proposed customer database will enable Nxt to approach and record all such leads to expand business to new customers. It may buy in lists, or search through the Official Journal of the European Union for available opportunities.

The limited B2C service is for secure items only (such as legal documents, travel tickets and currency). As these are profitable, the expansion of these services is highly desirable to Nxt. The proposed databases must be able to record these, but should also be available to marketing so that they can continue to grow and maintain that business. Because of the secure nature of this market segment, this data must be kept secure at all times with only certain personnel still allowed access.

Sales and Marketing

Nxt is well established in its core market. Nxt uses regional newspapers for advertising and relevant (target customer) trade press. It has a sales team which is organised regionally but with discreet staff working on Nxt Service and Nxt Secure (both companies' sales teams refer business to each other).

The company has had to innovate throughout its history, and has come a long way from hand sorting each item for delivery. However, whilst they recognise the need for innovation, they can sometimes be slow to respond – the management culture is very risk adverse and the marginal profitability means that The Nxt Day Group impose tough financial hurdles on any new investment.

Innovation

Nxt prides itself on its 95% on-time delivery, but it does have some parcel tracking issues – the company can usually tell a customer at the end of a working day whether or not a parcel was signed for, but it has to wait until individual couriers send in their signature delivery reports, or their own drivers return to the hubs with signature records. They do have a handwriting recognition technology which enters the data into a database, but it does make errors and most customers still rely on fax signatures for proof of delivery.

Nxt uses a system that tracks the progress of its vehicles, but it too has similar tracking issues – the drivers themselves tell their base where they are and when, but there is no independent verification of this, and occasionally, drivers can be creative with the truth – vehicles have been identified 'off route' on occasion, miles from where they were supposed to be, causing some customers significant inconvenience because of late collections or deliveries and also increasing operating costs for Nxt.

At this point in time, the company has come together to address the need for innovation because the market for businesses such as theirs is under threat from big European postal administrations and global courier companies…and they have to gain every possible business efficiency in order to remain competitive and still be able to offer reliable next-day delivery. The market is competitive, and margins can be described as "tight". They are keen to avoid the difficulties raised when companies are forced to cut their workforce, but recognise that this would likely be a reality if business looked elsewhere because they were unable to provide accurate up to date information on customer collections and deliveries.

They are keen to solve any logistics difficulties that can affect their customers' experience, because this affects their profits, particularly if a lost parcel is important enough to be newsworthy – sensitive data being lost in a parcel can have a direct, measurable impact on business for some weeks, not to mention being an expensive mistake – they have had occasions where the customer has sued for losses.

A suggestion has been made by the Chief Information Officer (CIO) to incorporate two new technologies that would go some way to solving the real logistics issues facing the company – one is a system by which lorries can be tracked by Global Positioning System (GPS) satellite as part of a vehicle telemetry system to improve management of their own fleet, and the other is a roll-out of Personal Digital Assistants (PDA's) throughout their own and sub-contractor courier networks. Deliveries could be signed for at the door with instant access to the time received, the sales force would have instant access to data they might need to secure a sale, or to refer a potential sale to the other network, and company management would be able to communicate instantly when there is a crisis in delivery – a closed motorway, natural disaster, or anything else that might prevent promised delivery schedules.

The CIO, which has always been a very important position, has not previously instigated change in a way that directly affects the day-to-day operation and involves drivers and hub personnel; and it will change the nature of how the company interfaces with its Information Technology (IT). This position is now viewed by the board as a key role, and the CEO and CFO are considering letting the CIO into the top-level company decisions, which is a departure from the past, where much of the IT operation was left to contractors. Where the CEO and CFO are somewhat fearful of the possible risks of the new technology, its smaller competitors have done limited roll-out of this technology, to some good effect.

Nxt knows that to keep its market-share, it will need to innovate, and is now also seeking guidance on how to improve its distribution network. It will review all of its business processes with a view to making necessary changes to its structure in consultation with staff and specialists to come up with a "best-fit" solution.

PDAs

So what is a PDA?

A PDA is a handheld computer used to capture customer signatures and scan barcodes upon delivery. This information is then transmitted in 'real time', using radio communication, from the customer's doorstep - improving the service level of tracked products delivered by vehicle.

Why PDAs?

Electronic handheld computers are an increasingly common part of the delivery experience for packet, express, tracked and signed for products.

Nxt must look to this new exciting technology in order to remain competitive and improve the existing products for customers. PDAs will allow Nxt to do this and help Nxt to compete. PDAs will provide customers with 'real time' tracking and confirmation of delivery. This information is collected and stored in a data base, allowing Nxt to assess efficiency, detect persistent problems with particular deliveries and sub-contractors, and can he used by marketing to detect new sales opportunities.

What does it mean for Nxt?

PDAs will provide Nxt delivery drivers with time saving device that will materially change the way they work. This will replace the paper / pen signature process and speed up deliveries, but may meet with some resistance. PDAs can be used as a mobile phone, keeping drivers in touch with their office and increasing their safety when out on delivery. They will be able to interact with barcodes to make deliveries more timely, and lost parcels less likely. They can also be used as a satellite navigation tool to improve navigation and help Nxt get to customers quicker. This flexibility means that should a route become blocked or another route is more efficient, a decision about a change is made in real time. It also means that courier movement can be tracked, and therefore it needs to be presented as a work-enhancing device.

What about the 75% of items delivered by couriers?

Nxt wants to be able to roll out PDAs to all courier service providers based on a successful deployment to Nxt own staff.

Vehicle Telemetry

So what is Telemetry?

Telemetry is all about sharing information. Vehicles automatically record and send information, straight back to a computer in the hub where the information can be viewed. It shows in real time where all vehicles are and where they have been.

The data could include such things as when a van stops and starts, its location and a vehicle's arrival time at the hub or customer compared to the scheduled time. With telemetry, hubs will be able to see where all its vehicles are, tracking via vehicle number, what routes they are doing identified by recognised named routes, how they are being driven by measuring speed and braking, and if there are any problems en route for example traffic or flooding. This helps Nxt to improve driver safety and get the most from the vehicle fleet. This information can be saved into the database, allowing Nxt to search for more efficient routes, and to identify places where couriers lose time.

Nxt would prefer a fully integrated system, meaning such contractors could also be included in telemetry benefits. However, with sub-contractors being paid a flat fee, it is less mission-critical for the moment. In the future, Nxt would like to handle a greater percentage of their traffic, meaning that all changes that benefit delivery efficiency will be more keenly felt.

Telemetry is also important in improving collection and delivery routes, and reducing fuel consumption by monitoring driving style. Driving information would automatically feed back to the database. Driving trends could be monitored automatically, flagging up unusual variations in speed, fuel usage, and fleet damage, all of which will help ensure that Nxt operates the most efficient routes that provide the service that customers want, and to help lower operating costs.

Why fit telemetry units?

Technological advances have improved the robustness and reliability of telemetry, tracking, communication and navigation systems.

Nxt must look to this new exciting technology in order to remain competitive and reduce its costs by making better use of its vehicles and drivers. TUs will allow Nxt to do this and help Nxt to compete. TUs will provide depot managers with 'real time' vehicle tracking, performance measurement, 2-way communications and route optimisation. Vehicle performance and routing information is collected and stored in a data base, allowing Nxt to assess efficiency, detect persistent problems with particular vehicles or drivers, and can he used by operations to increase the usage of Nxt's own fleet reducing reliance on sub-contractors.

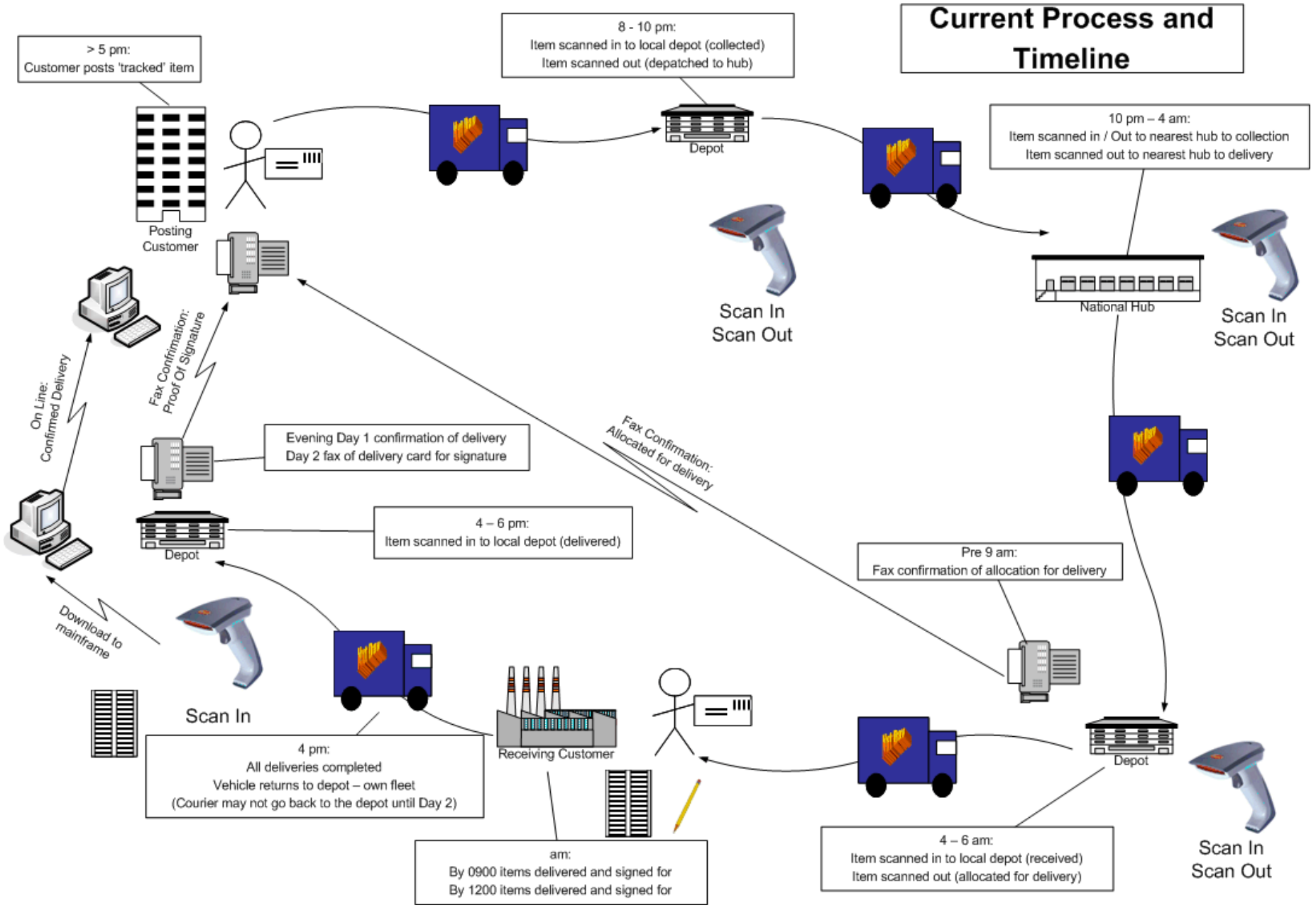
What does it mean for Nxt?

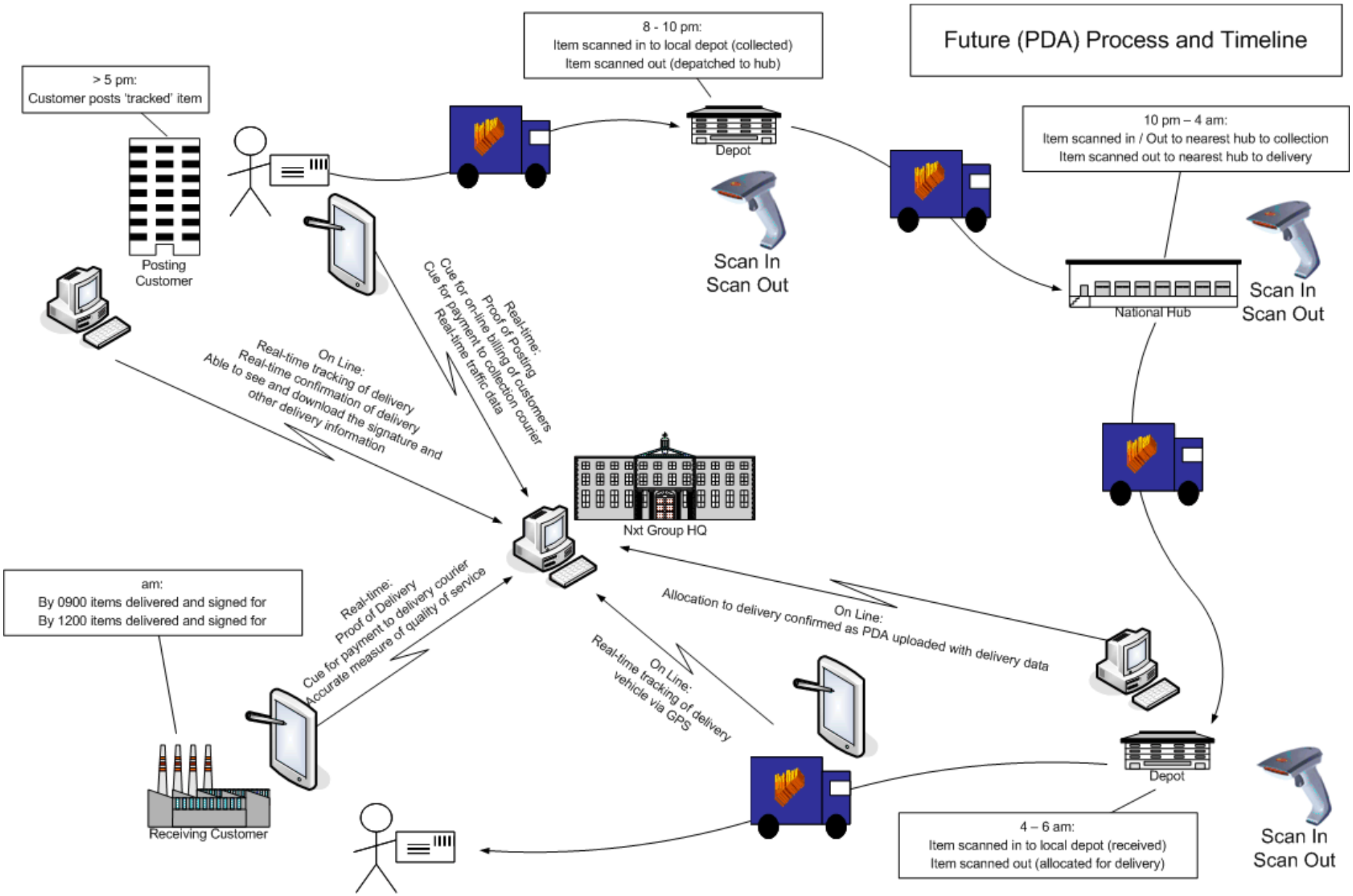
Telemetry will provide Nxt depots with real-time data that will materially change the way depots and drivers work. This will replace anecdotal information on the performance of

vehicles and drivers, increasing the efficiency and flexibility of the Nxt fleet, but may meet with some resistance from those who see it as a 'spy in the cab'. TUs can be used to keep drivers in touch with their office and thereby increase their safety when out alone. Built-in GPS technology automatically supplies information on the precise location of a vehicle. The telemetry system can calculate journey times and estimate arrival times to help keep customers informed The system captures detailed vehicle and driver performance data in real time including fuel consumption, driving style, vehicle performance etc as it is directly linked to the vehicle's engine management system. Like PDAs, TUs can also be used as a satellite navigation tool to improve navigation and help Nxt get to customers quicker. This flexibility means that should a route become blocked or another route is more efficient, a decision about a change is made in real time. It also means that courier movement can be tracked, and therefore it needs to be presented as a work-enhancing device.

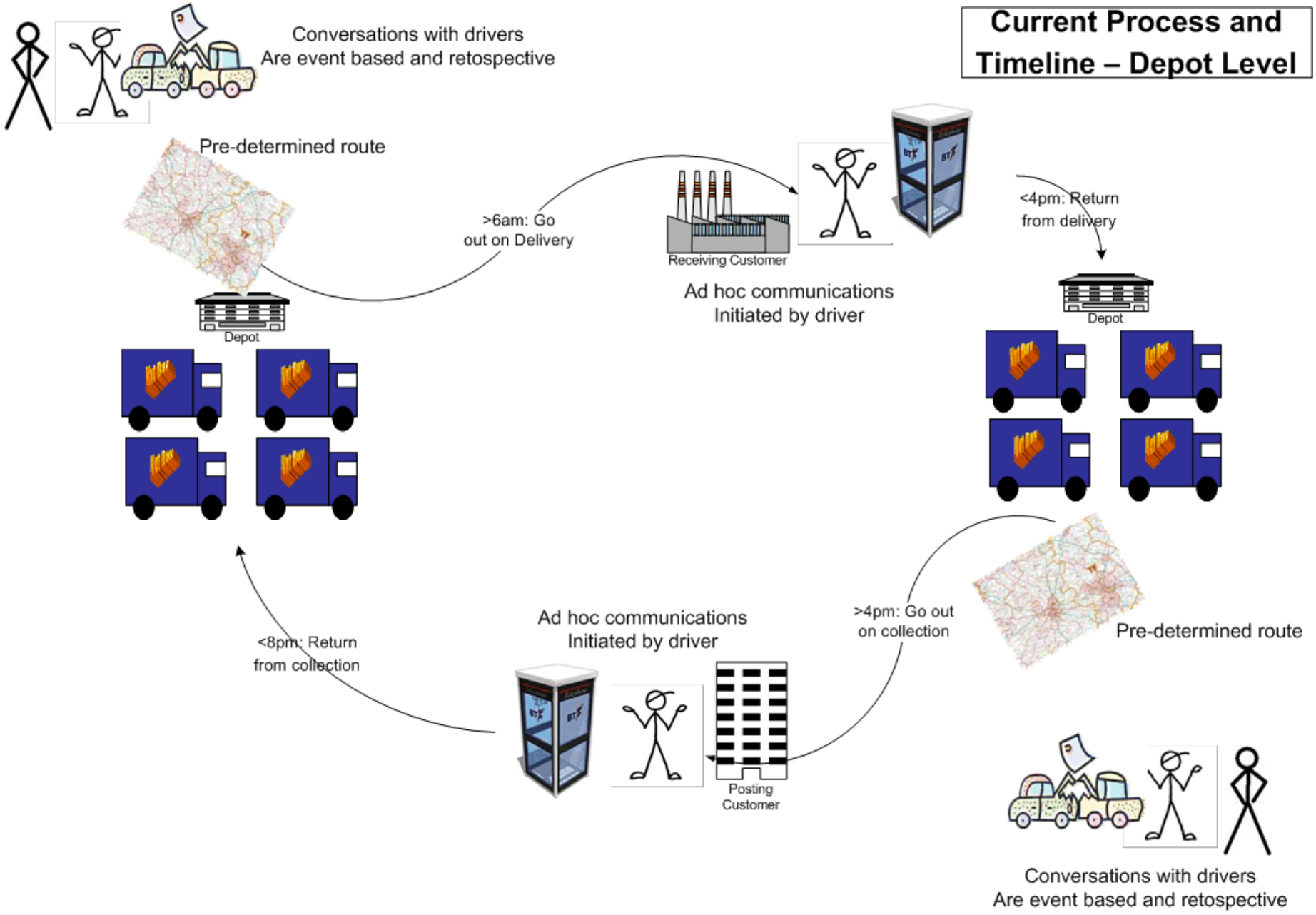
What about the 75% of items delivered by couriers' vehicles?

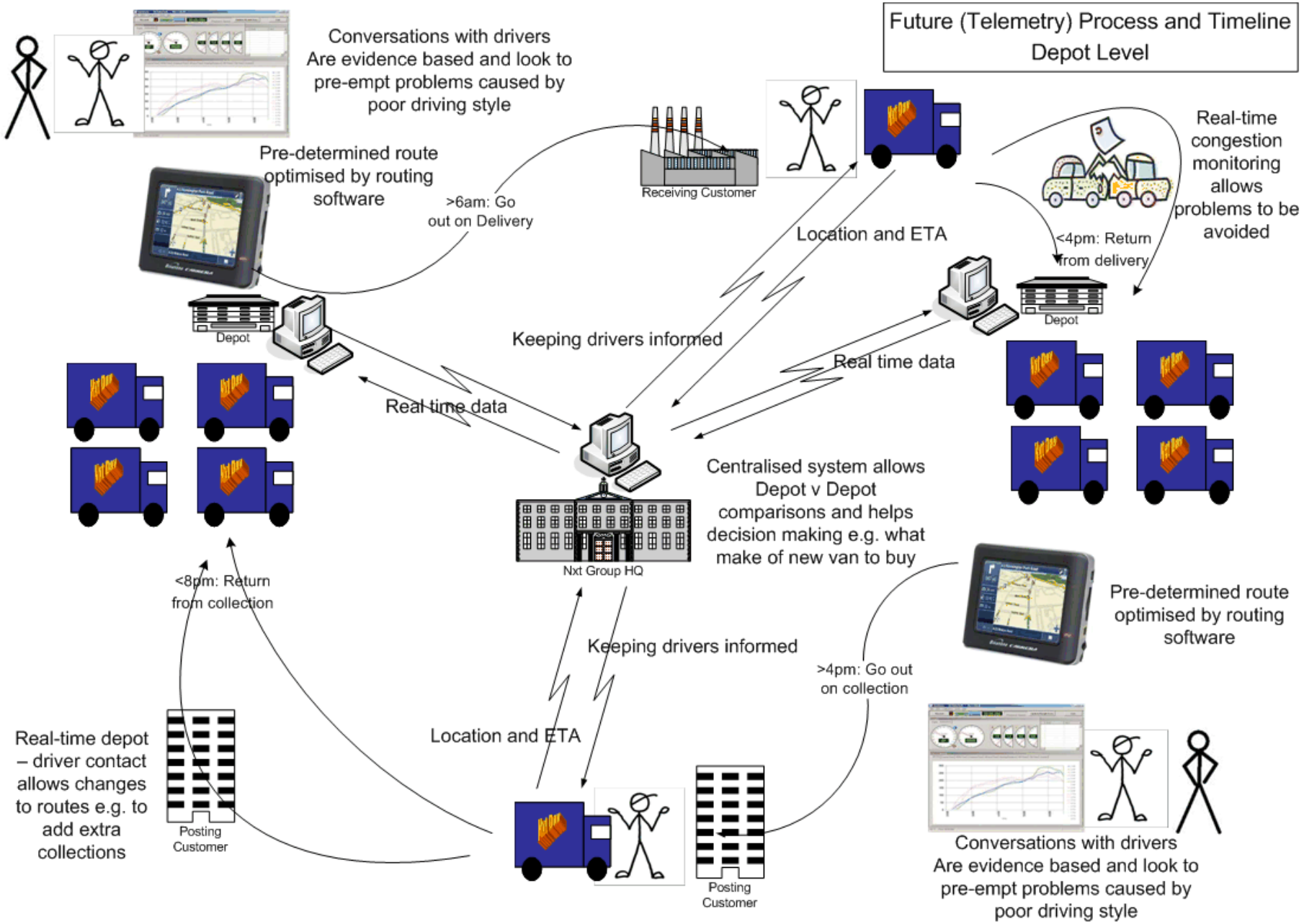
Nxt wants to be able to roll out TUs to all vehicles carrying its consignments based on a successful deployment to Nxt own staff but as sub-contractors are paid a flat fee, it is seen as less mission-critical for the moment.


Current Process and Timeline
> 5 pm:
Customer posts 'tracked' item
Posting Customer
8 - 10 pm:
Item scanned in to local depot (collected)
Item scanned out (depatched to hub)
Depot
Scan In
Scan Out
10 pm – 4 am:
Item scanned in / Out to nearest hub to collection
Item scanned out to nearest hub to delivery
National Hub
Scan In
Scan Out
On Line:
Confirmed Delivery
Fax Confirmation:
Proof Of Signature
Evening Day 1 confirmation of delivery
Day 2 fax of delivery card for signature
Fax Confirmation:
Allocated for delivery
Depot
4 – 6 pm:
Item scanned in to local depot (delivered)
Pre 9 am:
Fax confirmation of allocation for delivery
Download to mainframe
Scan In
4 pm:
All deliveries completed
Vehicle returns to depot – own fleet
(Courier may not go back to the depot until Day 2)
Receiving Customer
Depot
4 – 6 am:
Item scanned in to local depot (received)
Item scanned out (allocated for delivery)
Scan In
Scan Out
am:
By 0900 items delivered and signed for
By 1200 items delivered and signed for


Future (PDA) Process and Timeline
> 5 pm:
Customer posts 'tracked' item
Posting
Customer
8 - 10 pm:
Item scanned in to local depot (collected)
Item scanned out (depatched to hub)
Depot
Scan In
Scan Out
10 pm – 4 am:
Item scanned in / Out to nearest hub to collection
Item scanned out to nearest hub to delivery
National Hub
Scan In
Scan Out
On Line:
Real-time tracking of delivery
Real-time confirmation of delivery
Able to see and download the signature and
other delivery information
Real-time:
Proof of Posting
Cue for on-line billing of customers
Cue for payment to collection courier
Real-time traffic data
Nxt Group HQ
am:
By 0900 items delivered and signed for
By 1200 items delivered and signed for
Real-time:
Proof of Delivery
Cue for payment to delivery courier
Accurate measure of quality of service
On Line:
Allocation to delivery confirmed as PDA uploaded with delivery data
On Line:
Real-time tracking of delivery
vehicle via GPS
Receiving Customer
Depot
4 – 6 am:
Item scanned in to local depot (received)
Item scanned out (allocated for delivery)
Scan In
Scan Out

Current Process and Timeline – Depot Level




Future (Telemetry) Process and Timeline
Depot Level
Conversations with drivers
Are evidence based and look to
pre-empt problems caused by
poor driving style
Pre-determined route
optimised by routing
software
>6am: Go
out on Delivery
Receiving Customer
Location and ETA
Real-time
congestion
monitoring
allows
problems to be
avoided
<4pm: Return
from delivery
Depot
Keeping drivers informed
Real time data
Depot
Real time data
Nxt Group HQ
Centralised system allows
Depot v Depot
comparisons and helps
decision making e.g. what
make of new van to buy
<8pm: Return
from collection
Keeping drivers informed
>4pm: Go out
on collection
Pre-determined route
optimised by routing
software
Real-time depot
– driver contact
allows changes
to routes e.g. to
add extra
collections
Posting
Customer
Location and ETA
Posting
Customer
Conversations with drivers
Are evidence based and look to
pre-empt problems caused by
poor driving style